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# Paleomagnetism of Jurassic and Cretaceous Plutonic Rocks in the Sierra Nevada, California, and Its Significance for Polar Wandering and Continental Drift

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Directions of natural remanent magnetization (NRM) have been determined at thirteen sites in two intrusive igneous bodies in the western Sierra Nevada, California. Late Jurassic to Early Cretaceous potassium-argon ages of these rocks are: Guadalupe Mountains igneous complex, 136 m.y., and diorite of the Bucks batholith, 142 to 129 m.y. Normal and reversed NRM directions are found in both plutons. The mean paleomagnetic pole position corresponding to mean NRM directions at all sites after ac demagnetization is at 54°N 174°W, and the radius of the 95% confidence circle is 9.5°. This pole position is not significantly different from five previously published poles from widely separated North American rocks that range in age from 124 to 84 m.y. Such concordance strongly favors the geocentric dipolar configuration of the earth's magnetic field and further indicates that from about 138 to about 84 m.y ago polar wandering relative to North America did not occur. Comparison with similar paleomagnetic data from Australia and Africa suggests that apparent polar wandering during Mesozoic time was entirely due to continental drift and not to true polar wandering as an independent process.

## INTRODUCTION

The paleomagnetism of some of the Late Cretaceous granitic plutons that constitute the great central batholith of the Sierra Nevada, California, has been described by *Currie et al.* [1963] and *Grommé and Merrill* [1965]. Although those rocks yielded an apparently reliable paleomagnetic pole position that differs significantly from the present geomagnetic and geographic poles, difficulty was encountered at many of the sampling sites with unstable natural remanent magnetization (NRM) and excessive dispersion of NRM directions. The pole position found in that work is concordant with most other late Mesozoic poles obtained by other workers from North American rocks, but a very anomalous apparent pole position has been reported for rocks of similar age from the Franciscan formation in the California Coast Ranges [*Grommé and Gluskoter*, 1965] The purpose of the present investigation was to test the previous results from Sierra Nevada granites, to broaden the frame of reference for the results from the Franciscan formation, and to add to the still scarce paleomagnetic data for the late Mesozoic of North America.

## GEOLOGIC SETTING

A series of relatively isolated batholiths ranging in composition from gabbro to granodiorite crops out in the western foothills of the Sierra Nevada in northern California; similar batholiths also form the core of the Klamath Mountains in the northwestern part of the state. In contrast to the 85-m.y.-old granites that form the main part of the Sierra Nevada batholith, the potassium-argon ages of these rocks are grouped around 140 m.y (Late Jurassic according to *Kulp's* [1961] time scale). The age, stratigraphic relations, and tectonic environment of these earlier granitic rocks have been summarized by *Curtis et al.* [1958].

Three of these plutons, all in the foothills of the Sierra Nevada, were selected for paleomagnetic study because they are accessible and well exposed, they had been mapped geologically, and at least some radiometric ages were available for them. These are the Guadalupe Moun-

tains igneous complex previously informally referred to as the Guadalupe Mountain pluton or quartz monzonite by *Curtis et al.* [1958], the Guadalupe igneous complex by *Best* [1961, 1963], and the Guadalupe complex by *Bowen* [1963]; the Merrimac batholith, previously referred to as the Merrimac pluton or granodiorite by *Hietanen* [1951]; and the Bucks batholith, referred to as the Belden granodiorite by *Curtis et al.* [1958], and the Bucks pluton or granodiorite by *Hietanen* [1951]. The Guadalupe Mountains igneous complex is east of Merced in Mariposa County, and the Merrimac and Bucks batholiths are between Oroville and Quincy in Plumas County (*Burnett and Jennings* [1962] and G. D. Turner, personal communication, 1963). Previously published potassium-argon ages from these rocks are 136 m.y. for quartz monzonite in the Guadalupe Mountains igneous complex, and 129 m.y. for granodiorite in the Bucks batholith; these ages have been recalculated from data given by *Curtis et al.* [1958] using the decay constants in Table 1. Newly determined ages, not previously published, have been obtained for the Merrimac and Bucks batholiths through the courtesy of J. F. Evernden and are given in Table 1 together with the relevant analytical data. The sample locations are given in the appendix, and the ages will be discussed more fully below.

## Experimental Methods

*Sampling scheme.* Approximately ten oriented cores were obtained at each of fourteen sampling sites by means of the procedure described by *Currie et al.* [1963]. The spacing between cores ranged from ½ to 150 meters, depending on the extent of exposed rock. From two to four specimens were cut from each core. On the basis of the behavior of NRM directions during demagnetization, three of the sites (3, 10, and 11) were each divided into two parts, and each part was subsequently treated as a single site.

In these rocks the dispersion of NRM directions among specimens cut from a single core is as great as the dispersion among a group of cores from a single site. (The same condition had been found in granites of Yosemite Valley and Sonora Pass [*Currie et al.*, 1963; *Grommé and Merrill*, 1965].) Therefore, in calculating the mean NRM direction for each site we have assigned unit weight to each specimen rather than to each core. This is in contrast to the procedure usually used in dealing with NRM of lavas, where the within-core dispersion is commonly much less than

TABLE 1. Potassium-Argon Ages of Mineral Concentrates from the Bucks and Merrimac Batholiths, Plumas County, California (J. F. Evernden, personal communication, 1964)

Decay constants: $\lambda_e = 0.585 \times 10^{-10}$ yr$^{-1}$; $\lambda_\beta = 4.72 \times 10^{-10}$ yr$^{-1}$. See Appendix for sample localities shown in part on Figure 2.

| Rock Type | Mineral | Sample No. | Sample Weight, g | K, wt. % | $A^{40}_{at}$, % | Age, $10^6$ years |
|---|---|---|---|---|---|---|
| | | *Bucks Batholith* | | | | |
| Quartz diorite | Biotite | KA1436 | 0.3499 | 7.031 | 10 | 132 |
| | Hornblende | KA1437 | 1.167 | 0.5945 | 50 | 136 |
| Hornblende diorite | Hornblende | KA1442 | 1.158 | 0.4515 | 32 | 143 |
| | Hornblende | KA1449 | 2.451 | 0.4512 | 14 | 142 |
| Quartz diorite | Biotite | KA1432 | 0.4116 | 7.607 | 6 | 128 |
| | Hornblende | KA1433 | 0.6273 | 0.6461 | 17 | 142 |
| Granodiorite | Biotite | KA1440 | 0.3507 | 7.485 | 6 | 129 |
| | Hornblende | KA1441 | 1.372 | 0.5770 | 24 | 132 |
| | | *Merrimac Batholith* | | | | |
| Granodiorite | Biotite | KA1438 | 0.3177 | 7.335 | 24 | 131 |
| | Hornblende | KA1439 | 1.319 | 0.5817 | 6 | 129 |

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the between-core dispersion [*Cox and Doell*, 1960; *Doell and Cox*, 1965].

*Stability of remanent magnetization.* The NRM's of all specimens were measured as soon as possible after collection and again after a period of from 7 to 16 months. During this period the specimens were stored with their NRM vectors at large angles to the earth's field in the laboratory, a procedure that produces maximum dispersion in groups of specimens with highly unstable NRM. For the Cretaceous granitic rocks near Sonora Pass this effect was found to be very great at all sites [*Grommé and Merrill*, 1965]. In the present work, of the 11 sites tested in this way only 1 showed an increase in dispersion that was statistically significant according to the test of *McElhinny* [1964]. This site was in the Merrimac batholith, which will be discussed in a later section.

*The ac demagnetization.* To test further the stability of the NRM and to remove relatively softer secondary components of magnetization, ac demagnetization was performed on specimens from all sites. The demagnetizing apparatus used was the same as that described by *Currie et al.* [1963]; this device rotates ('tumbles') the specimen around three axes simultaneously to achieve systematic demagnetization in all directions. Many workers [e.g., *Irving et al.*, 1961] have found that a specimen demagnetized in this way often acquires a small and apparently randomly directed 'stray' remanent magnetization; this effect is most noticeable at alternating fields greater than about 200 oe. *Doell and Cox* [1967] have analyzed these spurious magnetizations and find that they are not randomly directed but tend strongly to be parallel to the innermost tumbler axis. Doell and Cox conclude that the spurious magnetizations are anhysteretic in character in that they depend partly on the magnitude and orientation of the steady ambient field. Exactly the same behavior has been found in our demagnetization apparatus [*Grommé*, 1963]. For most rocks this effect is very small, and systematic variation of specimen orientation in the tumbler minimizes any possible systematic bias of NRM directions when the results for a group of specimens are averaged.

Because of these stray magnetizations, the optimum demagnetizing field for a given site is not always the field that gives minimum dispersion of NRM directions [*Irving et al.*, 1961] but may be somewhat greater. In some cases it is necessary to demagnetize progressively three or more specimens from one site and average the directions at each increment of demagnetizing field. The optimum ac field can be chosen by comparing increase in dispersion with further systematic change in the mean direction. We have followed this procedure in the present work and find that either the minimum in dispersion coincides with the end of systematic movement of the mean direction or any further systematic direction change is made insignificant by the increase in dispersion. These results are shown below in the figures accompanying the discussion of data from individual sites.

### Guadalupe Mountains Igneous Complex

*Geology.* Figure 1 is a geologic map of the Guadalupe Mountains igneous complex (simplified from *Best* [1961]), showing the approximate location of the specimen dated by *Curtis et al.* [1958] and our paleomagnetic sampling sites 1, 2, and 3. The layering in the gabbro comprising the western part of the complex is defined by igneous lamination (i.e., preferred orientation of mineral grains) and by rhythmic variation in grain size and relative abundance of constituent minerals. This layering is interpreted as being the result of sedimentation, modified by convective motion, during crystallization of the rock from the magma [*Best*, 1963, pp. 250–251]. The gross asymmetry of the entire complex and the steep northeast dip of the layering in the gabbro have led *Best* [1963, p. 251] to suggest that the gabbro may have been tilted to the east, perhaps before consolidation of the remainder of the complex. Best has also pointed out that the apparent 'unconformity' between the layered gabbro and the base of the granite and granophyre (cf. the profile in Figure 1) suggests that most of the tilting, if real, would have occurred before solidification of the granite and that the amount of post-granite tilting is probably limited to about 10°.

The agmatite in the central part of the complex consists of a light-colored granitic matrix containing angular, sometimes close-fitting blocks of mafic rock quite similar to the massive meladiorite that occurs just to be west. This peculiar rock type is considered by *Best* [1963,

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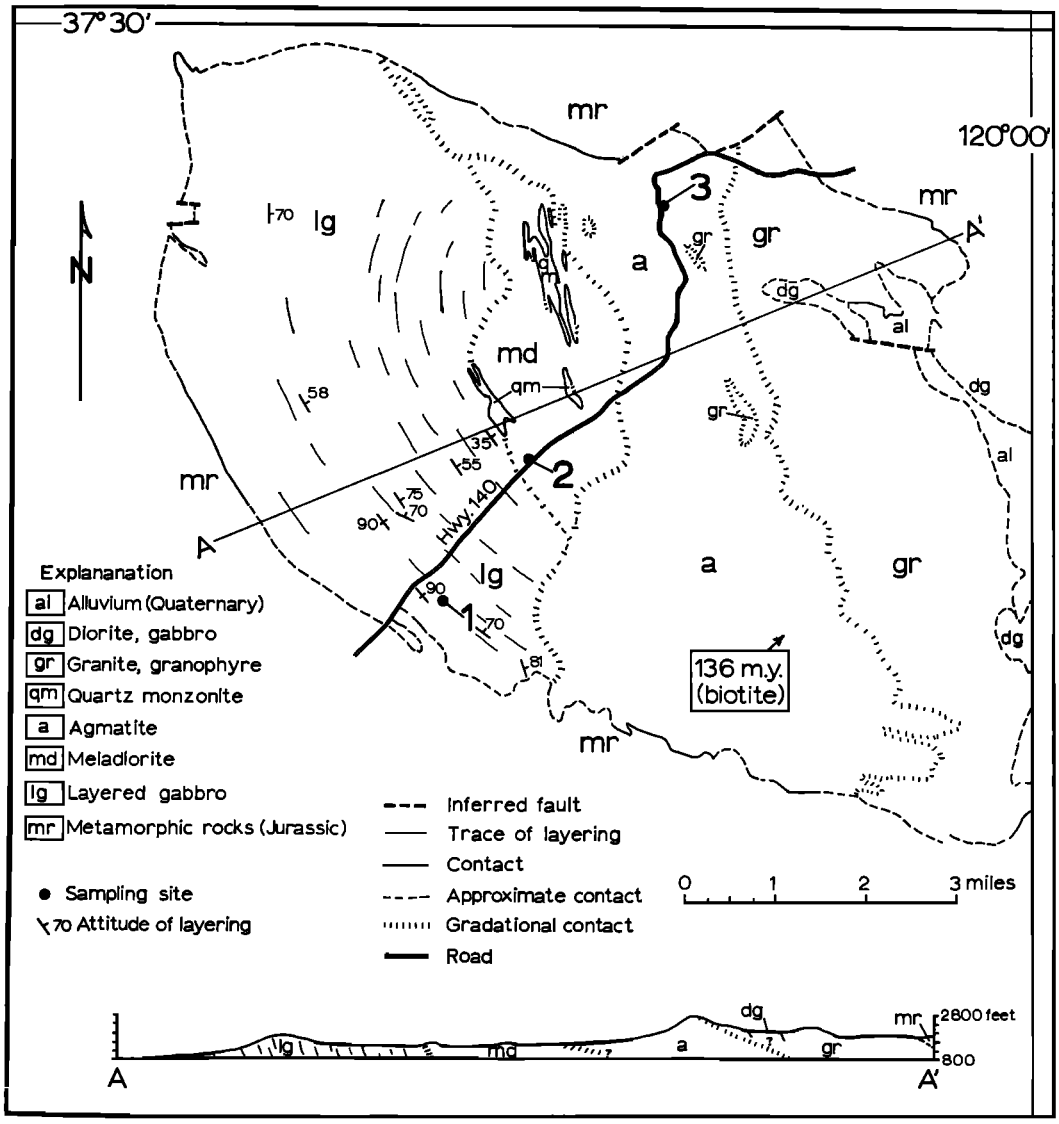



Fig. 1. Geologic map and cross section of the Guadalupe Mountains igneous complex and surrounding rocks, western Mariposa County, California, after *Best* [1961], showing paleomagnetic sampling sites 1–3 and the approximate location of the sample dated by *Curtis et al.* [1958].

p. 252] to be the result of collapse of the crystallized mafic roof of the complex into the still-molten granitic core. If these postulated movements (tilting and collapse) occurred after the rocks had cooled below the Curie temperatures of the enclosed ferromagnetic minerals, the movements might be expected to be reflected in systematic deviation of NRM directions in the gabbro and meladiorite and in random NRM directions among the mafic fragments in the agmatite.

*Natural remanent magnetization.* The ac demagnetization results for representative groups of specimens from four sites (1, 2, 3A, and 3B) in the Guadalupe Mountains igneous complex are shown in Figure 2. Complete results from the storage test and ac demagnetization are given in Table 2. For sites 1 and 2, in layered

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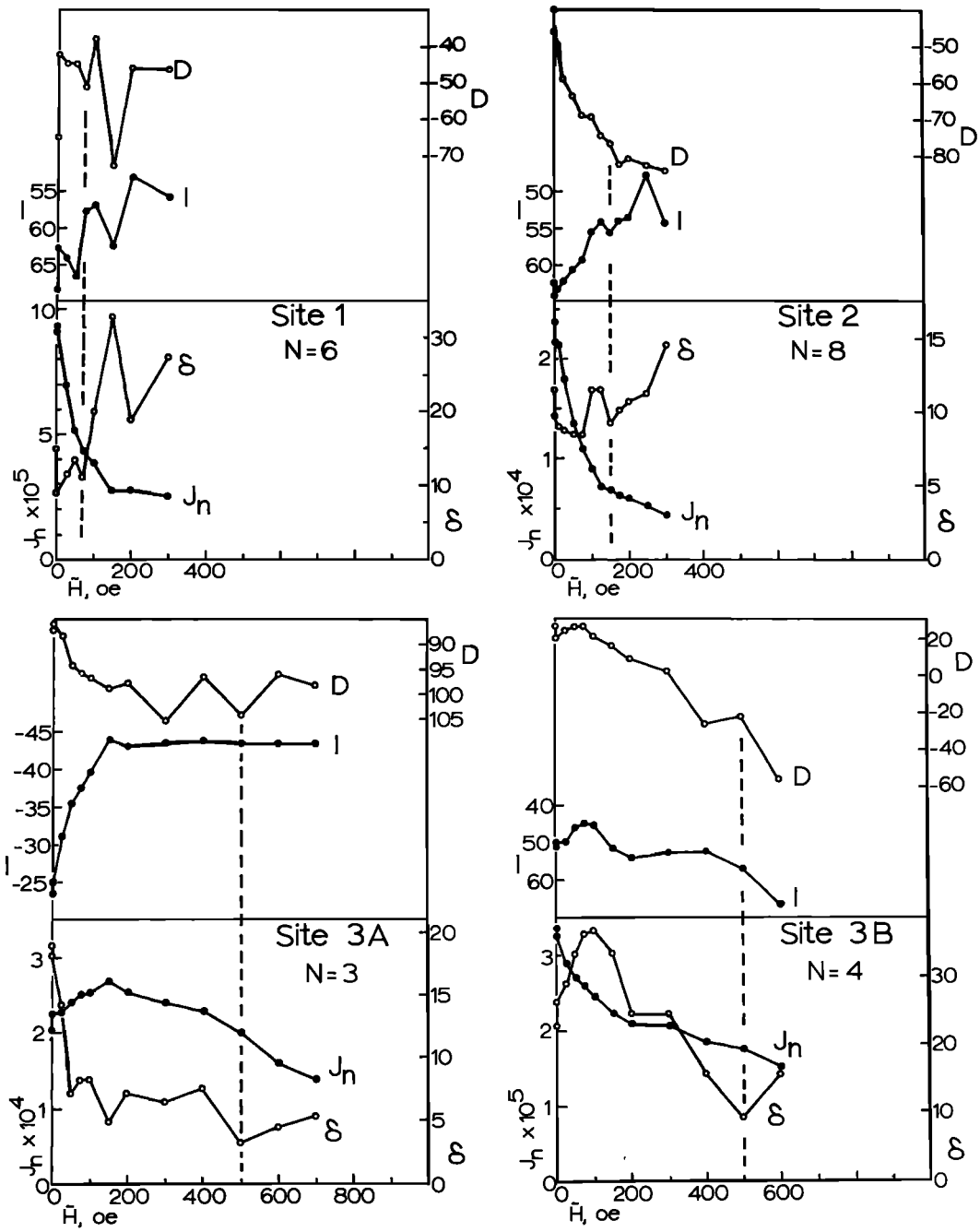



Fig. 2. The ac demagnetization results for NRM at all sites in the Guadalupe Mountains igneous complex. $D$ = mean declination, positive east of north. $I$ = mean inclination, positive downward. $J_n$ = mean intensity of NRM in emu/cc. $\delta$ = rms deviation of directions from the mean, in degrees [*Cox and Doell*, 1960]. $\bar{H}$ = peak demagnetizing field. $N$ = number of specimens. Heavy dashed line indicates optimum demagnetizing field. Pairs of points on the ordinates at $\bar{H} = 0$ represent measurements made immediately after collection and again after storage for several months. (In this and similar following diagrams the declination scales are proportioned relative to the inclination scales by the cosines of the average inclinations at the various sites.)

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TABLE 2. Mean NRM Directions at Four Sites in the Guadalupe Mountains Igneous Complex. (*a*) Measured after collection. (*b*) Remeasured after storage. (*c*) Measured after ac demagnetization.

| Site | $N$ | | $t$ | $\bar{H}$ | $R$ | $k$ | $\alpha_{95}$ | $D$ | $I$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 16 | *a* | 0 | ··· | 15.4790 | 28.8 | 7.0 | −51.5 | 66.1 |
| | | *b* | 15 | ··· | 15.7919 | 72.1 | 4.4 | −39.9 | 61.6 |
| | | *c* | ··· | 75 | 15.7196 | 53.5 | 5.1 | −50.8 | 62.1 |
| 2 | 27 | *a* | 0 | ··· | 25.7018 | 20.0 | 6.4 | −45.9 | 63.0 |
| | | *b* | 15 | ··· | 25.9643 | 25.1 | 5.7 | −52.3 | 62.5 |
| | | *c* | ··· | 175 | 26.1680 | 31.2 | 5.1 | −85.9 | 54.0 |
| 3A | 7 | *b* | 15 | ··· | 6.63804 | 16.6 | 15.3 | −84.4 | −26.9 |
| | | *c* | ··· | 500 | 6.99331 | 897. | 2.0 | 103.4 | −44.2 |
| 3B | 6 | *b* | 15 | ··· | 5.42172 | 8.6 | 24.2 | 5.8 | 55.9 |
| | | *c* | ··· | 500 | 5.93623 | 78.4 | 7.6 | −25.1 | 58.7 |

*Notes:*
$N$, number of specimens.
$t$, storage time, months.
$\bar{H}$, demagnetizing field peak, oersteds.
$R$, vector sum of $N$ unit vectors.
$k$, precision parameter estimate [*Fisher*, 1953].
$\alpha_{95}$, semiangle of 95% confidence cone [*Fisher*, 1953].
$D$, declination, positive east of north.
$I$, inclination, positive downward.

gabbro and meladiorite, respectively, the optimum demagnetizing fields are relatively low, and the effect of spurious remanence acquired during demagnetization can be clearly seen. At these two sites more than half of the NRM has coercivity less than 100 oe, which is the reason for the pronounced effect of these spurious magnetizations. During the early stages of demagnetization the mean directions of NRM move systematically away from the present direction of the earth's field, indicating that part of the softer NRM is of viscous origin. The sense of the NRM at sites 1 and 2 is normal.

Some of the cores from site 1 possessed abnormally high intensities of NRM (about $2 \times 10^{-3}$ emu compared with about $6 \times 10^{-5}$ emu for the remainder). During demagnetization of these particular specimens we found that the 'closure triangles' [*Graham*, 1955], which are plotted from measurements of NRM with a spinner magnetometer, became extremely large after demagnetization at higher fields (e.g. greater than 100 oe). This effect can be due to pronounced instability of NRM, but it can also arise from pseudo-remanent magnetization induced in an anisotropic specimen spinning in the earth's field [*Howell et al.*, 1958]. To check this possibility we measured the apparent susceptibility anisotropies of all specimens from site 1 in the plane normal to the axis of each core, using the method described by *Grommé and Merrill* [1965]. There is a strong correlation between abnormally high intensity of NRM, abnormally high susceptibility ($6 \times 10^{-3}$, contrasted with a normal value of $1 \times 10^{-4}$), and high anisotropy. Calling $P$ the ratio of maximum to minimum apparent susceptibilities [*Uyeda et al.*, 1963], we find that $P$ for the strongly magnetized specimens ranges from 1.17 to 2.50. $P$ for the remaining specimens could not be determined directly owing to the low susceptibility, but it is probably not much greater than unity because ac demagnetization did not produce large closure triangles. For this reason we have not used NRM data from the strongly magnetized specimens from site 1, although before demagnetization their NRM directions did not differ significantly from the others. The susceptibility anisotropy is apparently due to concentration of ferromagnetic grains along the 'bedding planes' in the layered gabbro.

The directions of NRM in the stably magnetized specimens from site 3 are shown in

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Figure 3. These directions are peculiar in that they are distributed approximately in a plane that passes near the present field direction and actually contains the axial dipole field direction. The usual cause of such a planar distribution is superposition of a viscous magnetization, variable in intensity and directed along the present field, on the stable NRM, which is taken to lie at the farthest extremity of the planar distribution. This cannot be the case at site 3, however, since both extremities of the planar distribution lie beyond the present geomagnetic field direction; i.e., the present field direction is contained well within the planar distribution. Only the least stable specimens had NRM directions initially along the present field; these

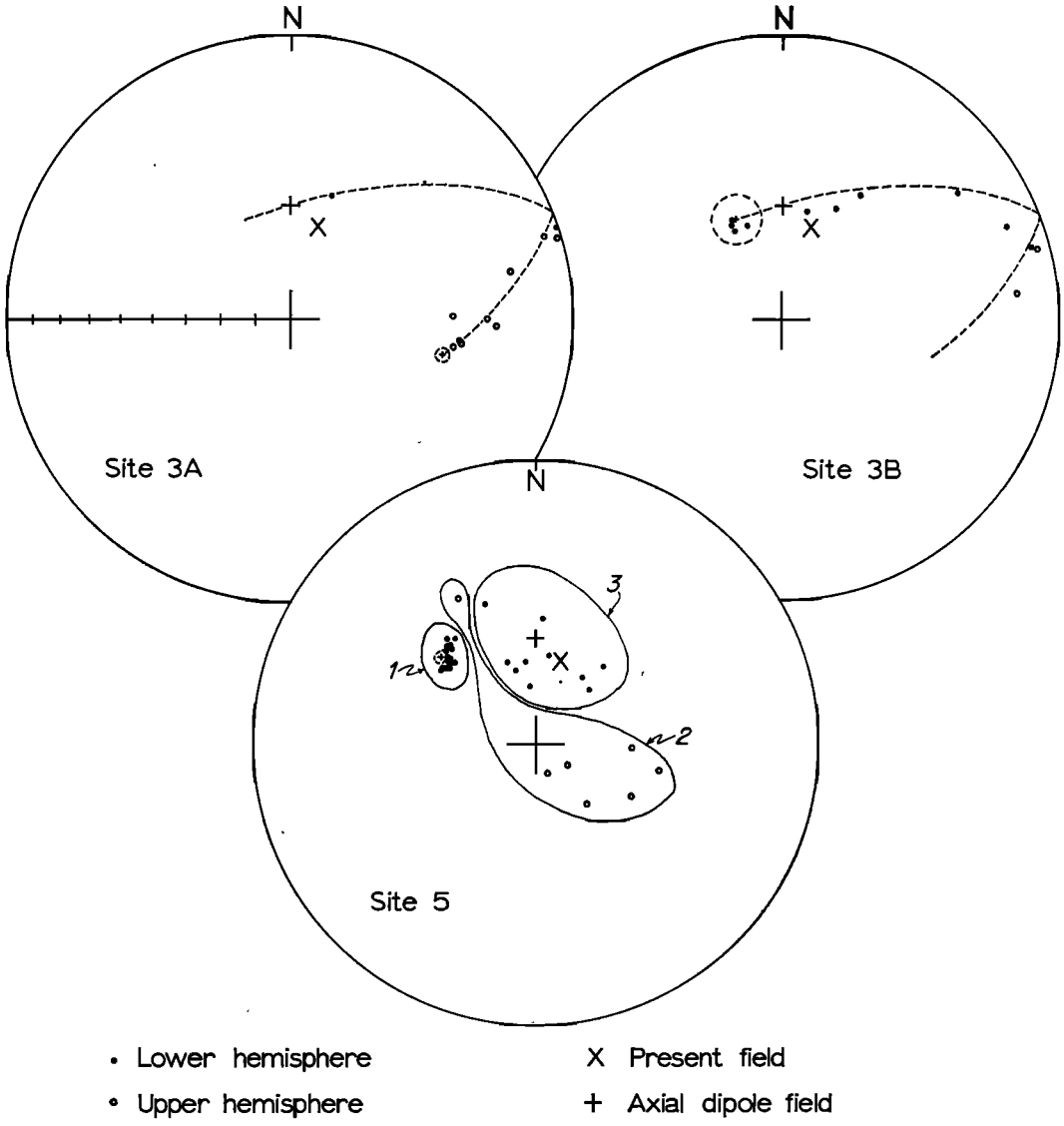



Fig. 3. Equal-area projections of north-seeking NRM directions as first measured, from site 3 in the Guadalupe Mountains igneous complex and site 5 in the Bucks batholith. Small plus sign indicates mean direction after ac demagnetization, and dashed circle is 95% confidence interval of this mean for each site (Tables 2 and 4). Great circle connects these mean directions at sites 3A and 3B.

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are few in number, did not pass the storage test, and are not represented in Figures 2 and 3. In the stably magnetized specimens the directions of NRM as first measured can be approximately correlated with location at the sampling site, which extends about 50 meters along a north–south roadcut. The reversely magnetized specimens are from nearer the north end and the normally magnetized specimens are mostly nearer the south end, but both directions have been found in adjacent specimens cut from the same core.

After ac demagnetization the NRM directions formed two distinct groups, normal and reverse (Figures 2 and 3); and on this basis site 3 has been divided into two parts. For both parts fairly high fields (500 oe) were required to produce minimum dispersion, as shown in Figure 2. Although normal and reversed NRM directions occur in adjacent specimens from the same cores before demagnetization, this is no longer true after demagnetization, and the boundary between sites 3A and 3B can be drawn clearly on the outcrop. The two parts of the site are about 30 meters apart, and each is only a few meters in extent.

At both sites in the agmatite, specimens were taken from several mafic fragments and from the granitic matrix. There is no systematic difference in magnetization directions between the two kinds; thus, the fracturing and collapse postulated by *Best* [1961, 1963] must have occurred while the rocks were still hotter than 600°C. This is hardly surprising since the granitic matrix must have been largely fluid during such a process.

## Merrimac Batholith

The Merrimac batholith is located about 20 miles northeast of Oroville in Plumas County (see *Burnett and Jennings* [1962]) and has been mapped and described by *Hietanen* [1951]. Only one well-exposed site was found in it, from which the dated sample (Table 1) was taken. (The location of this site is given in the appendix.) The age is about 130 m.y., or earliest Cretaceous according to *Kulp's* [1961] time scale.

Measurements of NRM at the site in the Merrimac batholith are summarized in Table 3. As calculated from the original measurements, the mean magnetization direction differed only slightly from the present field direction. After the specimens were stored for 14 months, the magnetization directions had become nearly random; *Fisher's* [1953] precision parameter $k$ decreased from 37 to about 4 (Table 3). Any stable component of NRM that might be carried by the rock is completely masked by a large soft magnetization. Because this soft component is so large, ac demagnetization leaves a random remanence that is larger than the stable NRM (if any exists), and we are forced to conclude that these rocks have no paleomagnetic value.

## Bucks Batholith

### *Geology*

The Bucks batholith is a complex three-lobed body ranging in composition from pyroxene diorite to granodiorite. It is located in Plumas and Butte counties about 35 miles northeast of Oroville. The approximate outline of the batholith is shown on the map by *Burnett and Jennings* [1962], and some aspects of its petrology were discussed by *Hietanen* [1951]. A more detailed map of the eastern (main) lobe and part of the central lobe is shown as Figure 4; it has been simplified from an unpublished map furnished by G. D. Turner (personal communication, 1963). In this figure are shown the locations of nine paleomagnetic sampling sites and

TABLE 3. Mean NRM Directions at One Site in the Merrimac Batholith
(*a*) Measured after collection. (*b*) Remeasured after storage.

| Site | $N$ | | $t$ | $R$ | $k$ | $\alpha_{95}$ | $D$ | $I$ |
|---|---|---|---|---|---|---|---|---|
| 4 | 35 | *a* | 0 | 34.0929 | 37.5 | 4.0 | −0.8 | 68.6 |
| | | *b* | 14 | 25.7454 | 3.7 | 14.8 | ... | ... |

Symbols as in Table 2.

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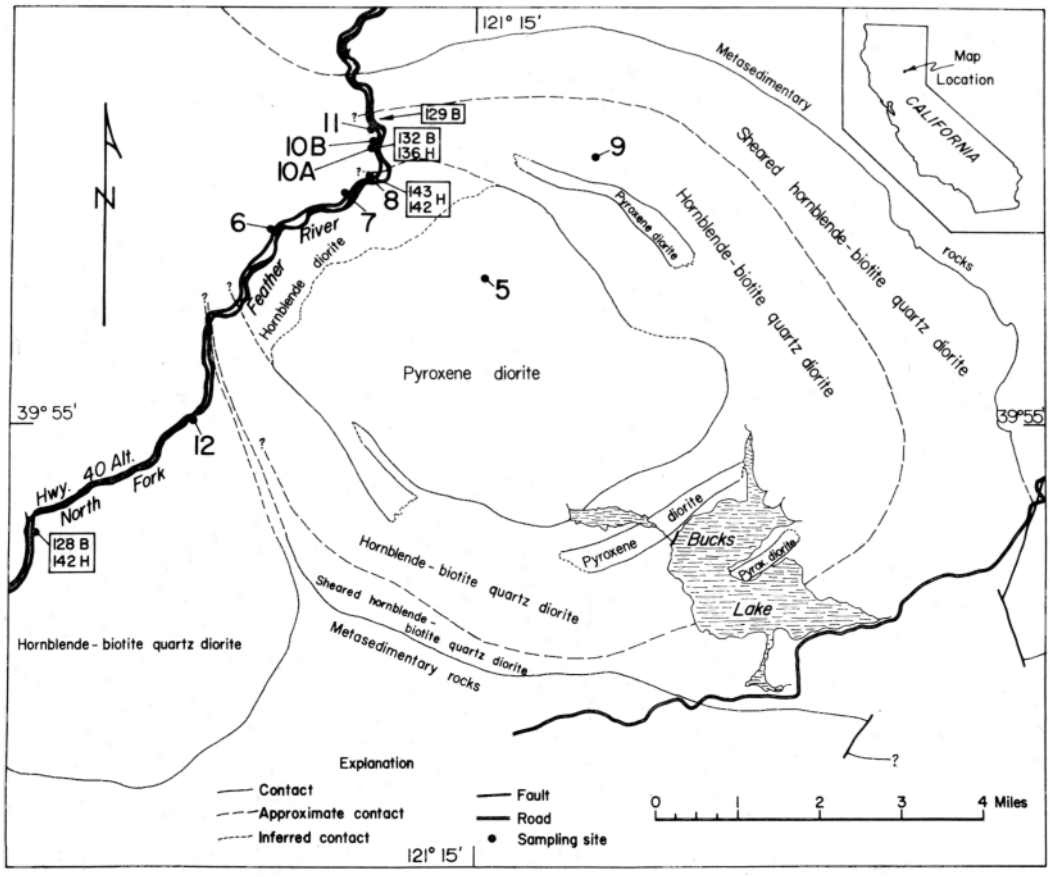



Fig. 4. Geologic map of the eastern part of the Bucks batholith after G. D. Turner (personal communication, 1963) and *Burnett and Jennings* [1962]. Locations of paleomagnetic sampling sites 5–12 are shown, together with the first six of the potassium-argon ages listed in Table 1 (B, biotite; H, hornblende). The 129-m.y. biotite age located near site 11 is recalculated from *Curtis et al.* [1958].

most of the new potassium-argon mineral ages listed in Table 1, as well as the 129-m.y. biotite age recalculated from *Curtis et al.* [1958].

Several aspects of the petrology of the Bucks batholith are pertinent to an interpretation of the paleomagnetic results to be described later. At site 5 (Figure 4) and elsewhere in the central pyroxene diorite distinct layering occurs; this layering is similar to that found in the Guadalupe Mountains complex except that in the Bucks batholith it is nearly horizontal wherever observed. If such layering is the result of gravitative settling of crystals within the partially molten pluton, we might infer that no post-consolidation tilting of the body has occurred. On the other hand, pervasive cataclasis in the outermost shell of the eastern lobe (the sheared hornblende-biotite quartz diorite shown in Figure 4) suggests post-consolidation movement.

Knowledge of the relative ages of the composite parts of the Bucks batholith would help to resolve its tectonic history and is relevant to interpretation of the paleomagnetic data. Where exposed in the Feather River canyon the contact between quartz diorite and hornblende diorite is gradational, whereas between sites 5 and 9 the quartz diorite appears on a small scale to be intrusive into the pyroxene diorite. The hornblende diorite exposed in the Feather River canyon contains fresh hornblende but strongly altered relict pyroxene, and the plagioclase is also altered; this alteration is in clear contrast to the fresh appearance under

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the microscope of the quartz diorite and the pyroxene diorite. The hornblende diorite thus appears to have been originally pyroxene diorite thus appears to have been originally pyroxene diorite like that at site 5, and the alteration of pyroxene and plagioclase and the formation of fresh hornblende may be evidence of contact metamorphism, resulting perhaps from later intrusion of the adjacent quartz diorite.

The potassium-argon ages do not contradict this interpretation of the field evidence, but they do not offer it much support. No age could be determined from the pyroxene diorite, owing to the lack of suitable minerals, and no fresh biotite was found in the hornblende diorite that could be dated to confirm the hornblende ages at site 8 (Figure 4). The difference between the hornblende ages at site 8 and site 10A may not be geologically significant; according to the above hypothesis of contact metamorphism, they should be the same, but some or all of the hornblende in the hornblende diorite (i.e. at site 8) may be primary (magmatic) in origin. Interpretation of the biotite ages is somewhat uncertain in view of the discordance between biotite and hornblende ages obtained southwest of site 12 in the central lobe of the batholith (Figure 4). The hornblende ages are probably more reliable because it has now been clearly shown that the argon retentivity of hornblende is much greater than that of biotite [*Hart*, 1964].

For the eastern lobe of this batholith to have crystallized in place as a single unit would have required rather unusual conditions, since the zoning of rock types is the reverse of what would normally be expected if the pluton had crystallized inward from the top and sides. The simplest hypothesis is that it represents a composite intrusion, as was already implied in the preceding discussion.

The potassium-argon mineral ages from the eastern lobe of the Bucks batholith range from 143 to 129 m.y. (Table 1), thus spanning the Jurassic-Cretaceous boundary according to *Kulp*'s [1961] time scale.

*Natural Remanent Magnetization*

The results of the storage tests and ac demagnetization of the NRM at nine sites in the Bucks batholith are summarized in Table 4. During storage times ranging from 7 to 16 months, the NRM directions at all sites remained essentially stable. Because at some sites in these rocks the NRM behaves in a complex manner during ac demagnetization, the sampling sites will be discussed in several groups.

*Site 5.* Site 5 is at an elevation of 1650 meters, in the only well-exposed part of the pyroxene diorite that forms the core of the Bucks batholith. The NRM directions, shown in Figure 3, fall into three distinct groups: group 1 has normal polarity and small dispersion, group 2 has reversed polarity and large dispersion, and group 3 has normal polarity and moderate dispersion but is very close to the present field direction. These divisions correspond only roughly to the distribution of samples over the outcrop; group 1 is somewhat separate from the rest but groups 2 and 3 overlap in position.

The ac demagnetization results for group 1 are shown in Figure 5. Almost no soft magnetization is present in these samples, and all the stable NRM has coercivity greater than 1000 oe, a remarkably high value. The NRM in group 2 specimens has a similarly high coercivity, and, although some soft magnetization is present, ac demagnetization did not reduce the large dispersion of directions or the apparent 'streaking' toward group 1. In the specimens of group 3, the NRM is unstable and more than half is removed by an ac field of 75 oe; this fact combined with the position of the mean direction indicates that the NRM of group 3 is nearly all of viscous origin. For these reasons, the only specimens that are considered to give reliable paleomagnetic results are those in group 1.

*Sites 6, 7, and 8.* These three sites are in hornblende diorite and are located in the canyon of the Feather River at an elevation of about 610 meters. The NRM's at sites 6 and 7 have reversed polarity but NRM at site 8 is normal. The ac demagnetization results for the three sites are shown in Figure 5. Relatively small amounts of viscous magnetization are present at all sites; they are removed by ac fields ranging from 50 to 250 oe. After demagnetization, the mean NRM direction for site 8 is almost exactly antiparallel to the mean NRM directions for sites 6 and 7. The intrinsic coercivity of NRM at these three sites is uniform and is low com-

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TABLE 4. Mean NRM Directions at Nine Sites in the Bucks Batholith
(*a*) Measured after collection. (*b*) Remeasured after storage. (*c*) Remeasured after ac demagnetization.

| Site | $N$ | | $t$ | $\bar{H}$ | $R$ | $k$ | $\alpha_{95}$ | $D$ | $I$ |
|---|---|---|---|---|---|---|---|---|---|
| 5 | 13 | *a* | 0 | ··· | 12.9782 | 551. | 1.8 | −45.7 | 54.3 |
| | | *b* | 15 | ··· | 12.9749 | 477. | 1.9 | −46.5 | 53.0 |
| | | *c* | ··· | 300 | 12.9852 | 811. | 1.5 | −47.2 | 52.6 |
| 6 | 17 | *a* | 0 | ··· | 16.5477 | 35.4 | 6.1 | 149.0 | −75.6 |
| | | *b* | 16 | ··· | 16.7953 | 78.2 | 4.1 | 147.0 | −73.7 |
| | | *c* | ··· | 100 | 16.7808 | 73.0 | 4.2 | 152.2 | −73.7 |
| 7 | 9 | *a* | 0 | ··· | 8.99159 | 951. | 1.7 | 150.1 | −72.0 |
| | | *b* | 8 | ··· | 8.98414 | 504. | 2.3 | 153.0 | −70.8 |
| | | *c* | ··· | 300 | 8.99681 | 2500. | 1.0 | 151.1 | −70.1 |
| 8 | 17 | *a* | 0 | ··· | 16.6595 | 47.0 | 5.3 | 0.2 | 70.7 |
| | | *c* | ··· | 200 | 16.8435 | 102.3 | 3.5 | −27.9 | 72.2 |
| 9 | 37 | *a* | 0 | ··· | 32.0217 | 7.2 | 9.4 | −58.8 | 73.9 |
| | | *b* | 15 | ··· | 32.1945 | 7.5 | 9.2 | −50.6 | 57.2 |
| | | *c* | ··· | 300 | 36.5117 | 73.7 | 2.8 | −69.9 | 74.6 |
| 10A | 4 | *a* | 0 | ··· | 3.99620 | 790. | 3.3 | −48.4 | 77.4 |
| | | *b* | 7 | ··· | 3.99637 | 826. | 3.2 | −42.3 | 77.1 |
| | | *c* | ··· | 150 | 3.99371 | 477. | 4.2 | −51.4 | 77.1 |
| 10B | 8 | *a* | 0 | ··· | 7.90840 | 76.4 | 6.4 | −61.3 | 73.8 |
| | | *b* | 9 | ··· | 7.88409 | 60.4 | 7.2 | −59.4 | 73.9 |
| | | *c* | ··· | 600 | 7.96122 | 180.5 | 4.1 | 141.4 | −71.1 |
| 11A | 4 | *a* | 0 | ··· | 3.99881 | 2525. | 1.8 | −62.6 | 76.4 |
| | | *b* | 8 | ··· | 3.99787 | 1410. | 2.5 | −56.4 | 76.7 |
| | | *c* | ··· | 50 | 3.99868 | 2277. | 1.9 | −59.0 | 76.7 |
| 11B | 7 | *a* | 0 | ··· | 6.85304 | 40.8 | 9.6 | −145.4 | 72.7 |
| | | *b* | 8 | ··· | 6.80381 | 30.6 | 11.1 | 167.5 | 78.9 |
| | | *c* | ··· | 800 | 6.94047 | 100.8 | 6.0 | 137.9 | −70.1 |

Symbols as in Table 2.

pared with groups 1 and 2 at site 5 but high compared with unstable NRM such as that at site 4 in the Merrimac batholith. From 40 to 50% of the NRM has coercivity greater than 250 oe, but less than 10% survives ac demagnetization at 1000 oe.

*Site 9.* This site is in a quarry in the outer (quartz diorite) part of the Bucks batholith at an elevation of 1860 meters. The NRM has normal polarity, but the directions as first measured are rather widely dispersed (Table 4). The ac demagnetization results are shown in Figure 6; most of the secondary magnetization is removed by a field of 100 oe and all of it by 250 oe. This secondary NRM is dispersed in direction and was apparently produced by lightning. The stable NRM has very high intrinsic coercivity; an alternating field of 1000 oe only reduces its intensity by about 30%.

*Sites 10A, 10B, 11A, and 11B.* These sites are also in the quartz diorite and are distributed over a total distance of 1 km in a north–south direction in the canyon of the Feather River at an elevation of 640 meters. As with site 3 in the Guadalupe Mountains complex, sites 10 and 11 have each been divided into two parts on the basis of polarity of NRM after ac demagnetization. The polarity of NRM after ac demagnetization is correlated with sample location within each site, and the 'A' parts, having normal polarity, are restricted to the south ends of each of the sites.

The NRM at sites 10 and 11 exhibits peculiar behavior and consequently has been investi-

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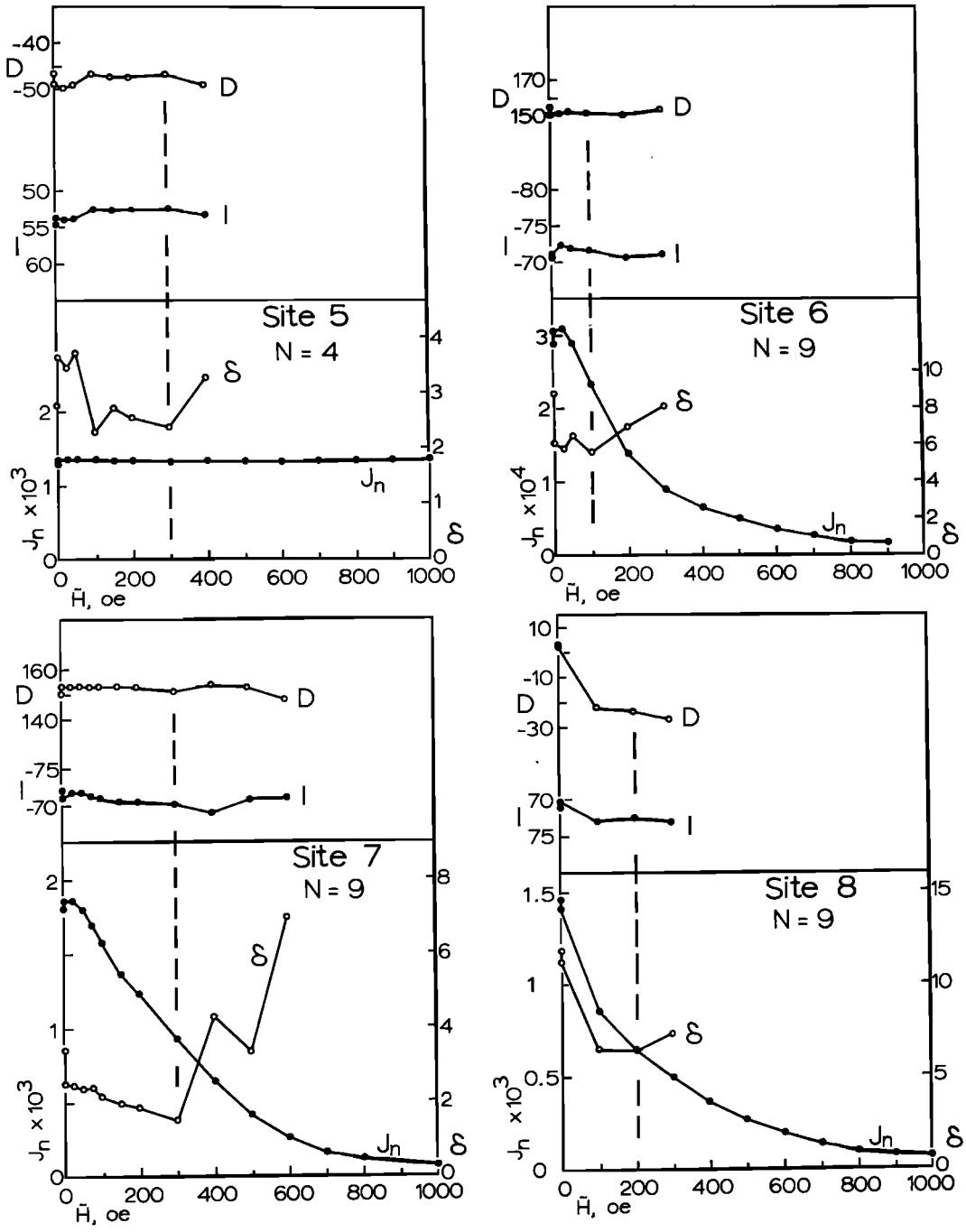



Fig. 5. The ac demagnetization results for NRM at sites 5, 6, 7, and 8 in the Bucks batholith. See Figure 2 for explanation of symbols.

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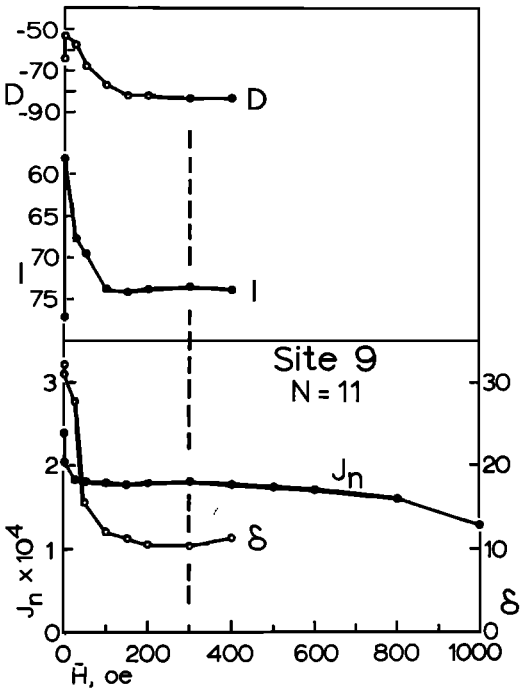



Fig. 6. The ac demagnetization results for NRM at site 9 in the Bucks batholith. See Figure 2 for explanation of symbols.

gated in detail. Directions prior to demagnetization are shown in Figure 7. At sites 10A and 11A they are normal and closely grouped. The directions at sites 10B and 11B are greatly scattered and are both normal and reversed; the directions at site 10B form a very crude planar distribution that is not related to the present or axial dipole field directions.

The ac demagnetization results for these four sites are shown in Figure 8. The polarities for 10A and 11A remain normal and the mean directions change only slightly. Minor secondary magnetization is removed by fields of 150 oe or less, and the stable NRM has coercivity distribution similar to that for sites 6, 7, and 8. In contrast, when specimens from sites 10B and 11B are demagnetized, the NRM directions, most of which were initially normal, become completely reversed and very closely grouped. These reversals are half complete at ac fields ranging from 100 to 300 oe and are complete at 800 oe.

In the upper part of Figure 7, two great circles are shown in the projections for site 10. One of these is defined by the normal and reversed directions of stable NRM at 10A and 10B, respectively, and the other represents the average path followed by NRM directions in specimens from site 10B during ac demagnetization and is roughly parallel to the planar distribution of initial NRM. Neither of these great circles contains the axial dipole or present field directions, and it is thus clear that the relatively soft normal component of NRM removed from the specimens of site 10B cannot be viscous magnetization due to the present field. This soft component also appears to differ slightly in direction from the stable NRM at site 10A.

Only one such great circle can be drawn for site 11: it connects the stable NRM directions for sites 11A and 11B (Figure 7). During ac demagnetization, NRM directions in specimens from site 11B tend very roughly to follow this great circle. Again it is clear that the relatively soft normal component of NRM in these specimens is not related to the present field; in this case, however, it appears to parallel the stable NRM at site 11A.

*Site 12.* The data from site 12 have proved to be of no paleomagnetic value. The intensities of NRM are very low, and the directions are scattered and unstable and do not respond to ac demagnetization in any consistent way. Site 12 was located in the bed of the Feather River, and it is worth noting that this site was the only one among those described above that was not located on an artificial exposure (i.e., on a road cut).

## Discussion

*Polarity of NRM.* Of a total of 10 main paleomagnetic sampling sites in the Guadalupe Mountains igneous complex and the Bucks batholith, 4 have normal NRM, 2 have reverse NRM, and 4 have mixed polarity. If the polarities of NRM all represent geomagnetic polarity at the times of initial magnetization of the rocks, a large number of geomagnetic field reversals (from three to five) is recorded in the Bucks batholith alone. On the other hand, self-reversal of NRM might have occurred at some of the sampling sites, for example, sites 3, 10B, and 11B. To pursue more fully the question of field reversal versus self-reversal we are deferring

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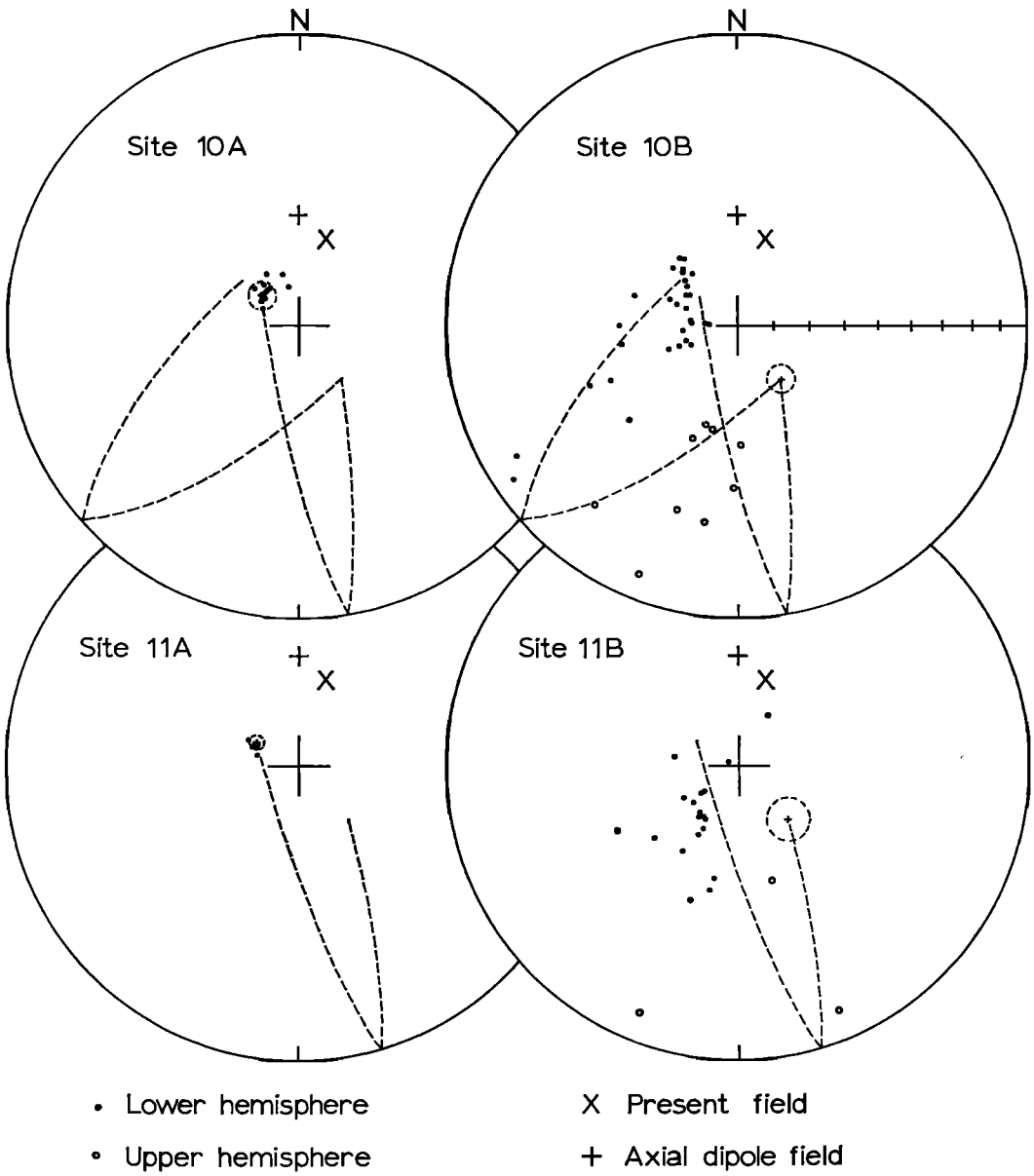



Fig. 7. Equal-area projections of north-seeking NRM directions as first measured, at sites 10 and 11 in the Bucks batholith. Small plus sign indicates mean direction after ac demagnetization, and dashed circle is 95% confidence interval of this mean for each site (Table 4). In upper projection, left-hand great circle represents average path followed by NRM directions at site 10B during ac demagnetization. Other great circles connect mean NRM directions after ac demagnetization, respectively, at sites 10A and 10B and sites 11A and 11B.

discussion of magnetic mineralogy and thermal demagnetization of NRM to a later paper.

*Site mean magnetization directions.* The mean NRM directions after ac demagnetization for the reliable sites or parts of sites in the Guadalupe Mountains igneous complex and the Bucks batholith are shown in Figure 9; this figure corresponds to the entries marked *c* in Tables 2 and 4. The corresponding paleomagnetic poles of the northern hemisphere calcu-

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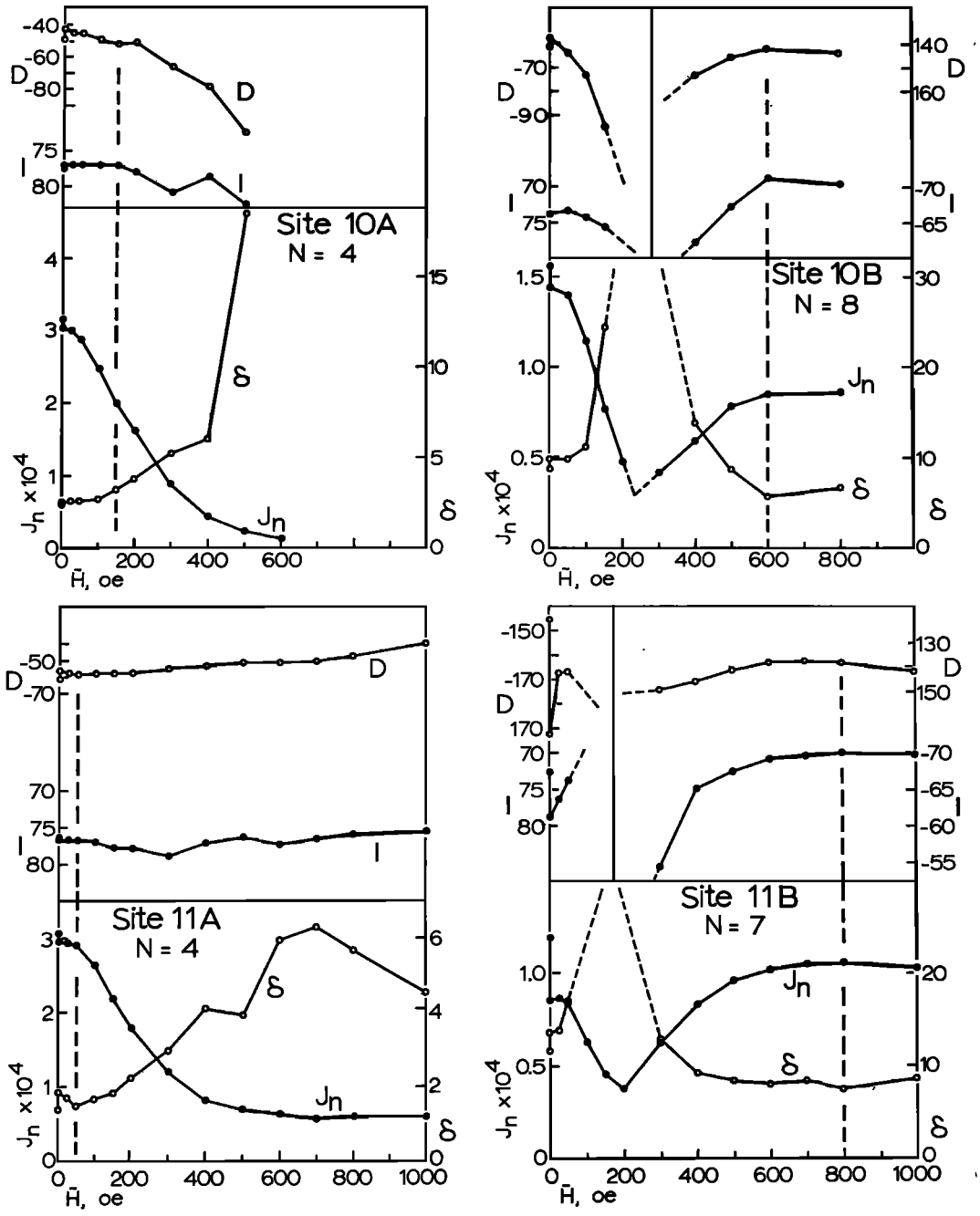



Fig. 8. The ac demagnetization results for NRM at sites 10 and 11 in the Bucks batholith. See Figure 2 for explanation of symbols.

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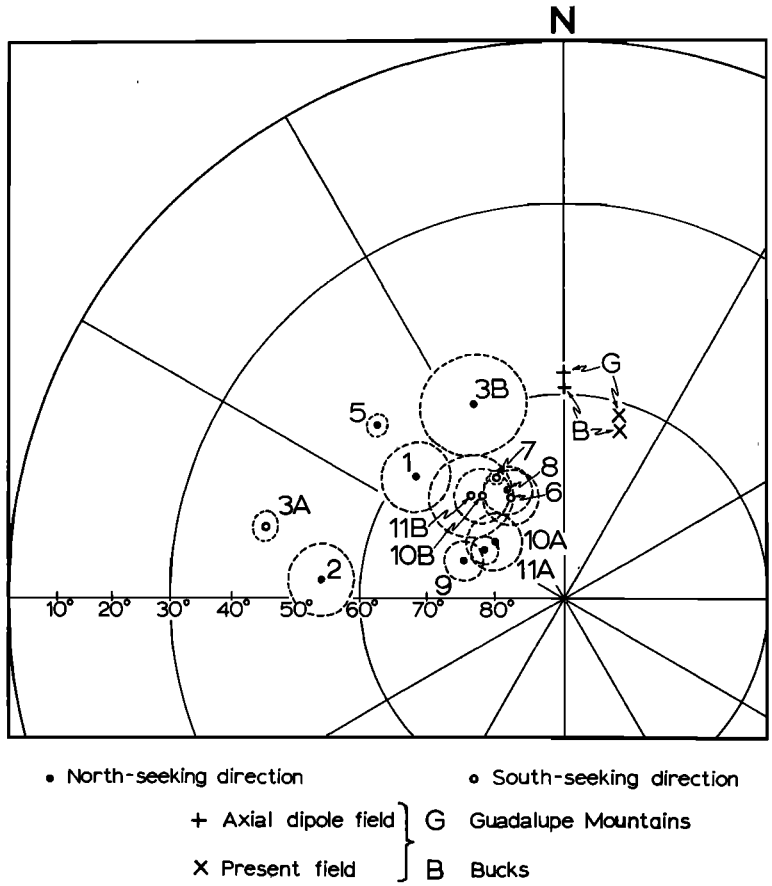



Fig. 9. Equal-area projection on lower hemisphere of mean NRM directions after ac demagnetization, at sites in the Gaudalupe Mountains igneous complex and the Bucks batholith. Dashed circles are 95% confidence intervals centered on the mean directions [*Fisher*, 1953].

lated on the usual assumption of a dipolar field are given in Table 5.

Several things are evident from Figure 9. First, the between-site dispersion is much greater for the Guadalupe Mountains complex than for the Bucks batholith. Second, the distributions of mean NRM directions in the two bodies, whether taken separately or lumped together, are not obviously Fisherian, which is probably the result of inadequate sampling with respect to geologic time. Third, the NRM direction at site 5 is well removed from all the others in the Bucks batholith, and these others are very tightly grouped. This agrees with the geological argument given above that the hornblende diorite (sites 6, 7, and 8) was magnetized at about the same time as the quartz diorite (sites 9, 10, and 11) rather than at the time the pyroxene diorite (site 5) was magnetized. Finally, the directions in the Guadalupe Mountains complex appear to be somewhat different from the directions in the Bucks batholith. This might indicate tilting of the Guadalupe Mountains complex, and the 10° rotation suggested by *Best* [1963] is in the right sense to produce this difference. To check this we plotted for both plutons the means of the site mean directions, and it turns out that the 95% confidence circle for the Guadalupe Mountains complex encloses the mean direction for the Bucks batholith. Thus, the resolution of the data is not sufficient to demonstrate rotation of the Guadalupe Mountains complex, and we shall assume that none occurred. In any event, it is clear that, if

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TABLE 5. Paleomagnetic Poles Calculated from Mean NRM Directions

After ac Demagnetization: Guadalupe Mountains Igneous Complex and Bucks Batholith.

| Site | $N$ | W Long. | N Lat. | $\delta_p$ | $\delta_m$ | Pole* |
|---|---|---|---|---|---|---|
| 1 | 16 | 184.7 | 51.5 | 6.2 | 7.9 | S |
| 2 | 27 | 183.4 | 23.0 | 5.0 | 7.1 | S |
| 3A | 7 | 196.0 | 25.6 | 1.6 | 2.5 | N |
| 3B | 6 | 196.3 | 70.3 | 8.4 | 11.3 | S |
| 5 | 13 | 206.2 | 51.9 | 1.4 | 2.0 | S |
| 6 | 17 | 153.4 | 63.7 | 6.8 | 7.6 | N |
| 7 | 9 | 165.4 | 66.0 | 1.5 | 1.8 | N |
| 8 | 17 | 158.0 | 64.9 | 5.5 | 6.3 | S |
| 9 | 37 | 160.1 | 43.5 | 4.6 | 5.0 | S |
| 10A | 4 | 152.8 | 51.5 | 7.3 | 7.9 | S |
| 10B | 8 | 166.6 | 60.3 | 6.3 | 7.2 | N |
| 11A | 4 | 154.9 | 48.5 | 3.3 | 3.6 | S |
| 11B | 7 | 170.3 | 58.6 | 9.0 | 10.4 | N |

*Notes:*

$N$, number of specimens.

$\delta_p$, $\delta_m$, semiaxes of 95% confidence oval, respectively parallel and perpendicular to the paleomagnetic meridian [*Cox and Doell*, 1960].

* S, south pole; N, north pole.

the layered gabbro was originally formed in a horizontal attitude, it was tilted before its magnetization.

## Paleomagnetic Significance

*Mean paleomagnetic pole.* The paleomagnetic poles for the northern hemisphere (site poles) in Table 5 have been combined to give the over-all mean pole position given in Table 6. Because the site mean NRM directions (and the corresponding site poles) apparently have a non-Fisherian distribution, there may be some doubt as to the statistical stability of the data. For this reason, in addition to the mean pole calculated from all sites, four mean poles calculated from subsets of the sites are also given in Table 6. The 95% confidence circles formally calculated for these mean poles [*Fisher*, 1953] all overlap one another, suggesting that the data are statistically stable. If the distribution of site poles is non-Fisherian, these 95% confidence circles have no rigorous statistical significance, so that statistical stability cannot be regarded as having been proved. Nevertheless, it appears that pole 1 in Table 6 (the mean of all site poles) is the most representative choice because it probably is least biased by effects of secular variation.

*Polar wandering relative to North America.* Pole 1 of Table 6 has been plotted in Figure 10 in order to compare it with other late Mesozoic paleomagnetic poles from North America and other continents. This pole has been assigned an age of 138 m.y., obtained by averaging the four new determinations for sites 8 and 10 (Table 1 and Figure 4) with the single determination for the Guadalupe Mountains igneous complex (Figure 1). In addition to this pole, five other paleomagnetic poles for the Cretaceous period of North America are shown in Figure 10; they have all been obtained from radiometrically dated igneous rocks. In the following discussion, numbers in parentheses are the serial numbers assigned by *Irving* [1964, pp. 306–310] in his compilation of paleomagnetic data. The pole from the Cretaceous granite of the Sierra Nevada, dated at 84 m.y., is from *Grommé and*

TABLE 6. Mean Paleomagnetic Poles Calculated from Data in Table 5

| Pole | $N$* | $k$ | $\alpha_{95}$ | W Long. | N Lat. | Notes† |
|---|---|---|---|---|---|---|
| 1 | 13 | 19.9 | 9.5 | 174.1 | 53.7 | All sites |
| 2 | 4 | 12.5 | 27.0 | 189.3 | 42.6 | Guadalupe Mountains sites only (1, 2, 3A, 3B) |
| 3 | 9 | 43.7 | 7.9 | 165.2 | 57.6 | Bucks sites only (5, 6, 7, 8, 9, 10A, 10B, 11A, 11B) |
| 4 | 4 | 31.0 | 16.8 | 175.4 | 55.0 | Bucks sites with high coercivity of NRM (5, 9, 10B, 11B) |
| 5 | 5 | 88.3 | 8.2 | 156.3 | 59.0 | Bucks sites with moderate coercivity of NRM (6, 7, 8, 10A, 11A) |

* $N$, number of site-poles. Other symbols as in Table 2.

† Indicate site poles included in each mean.

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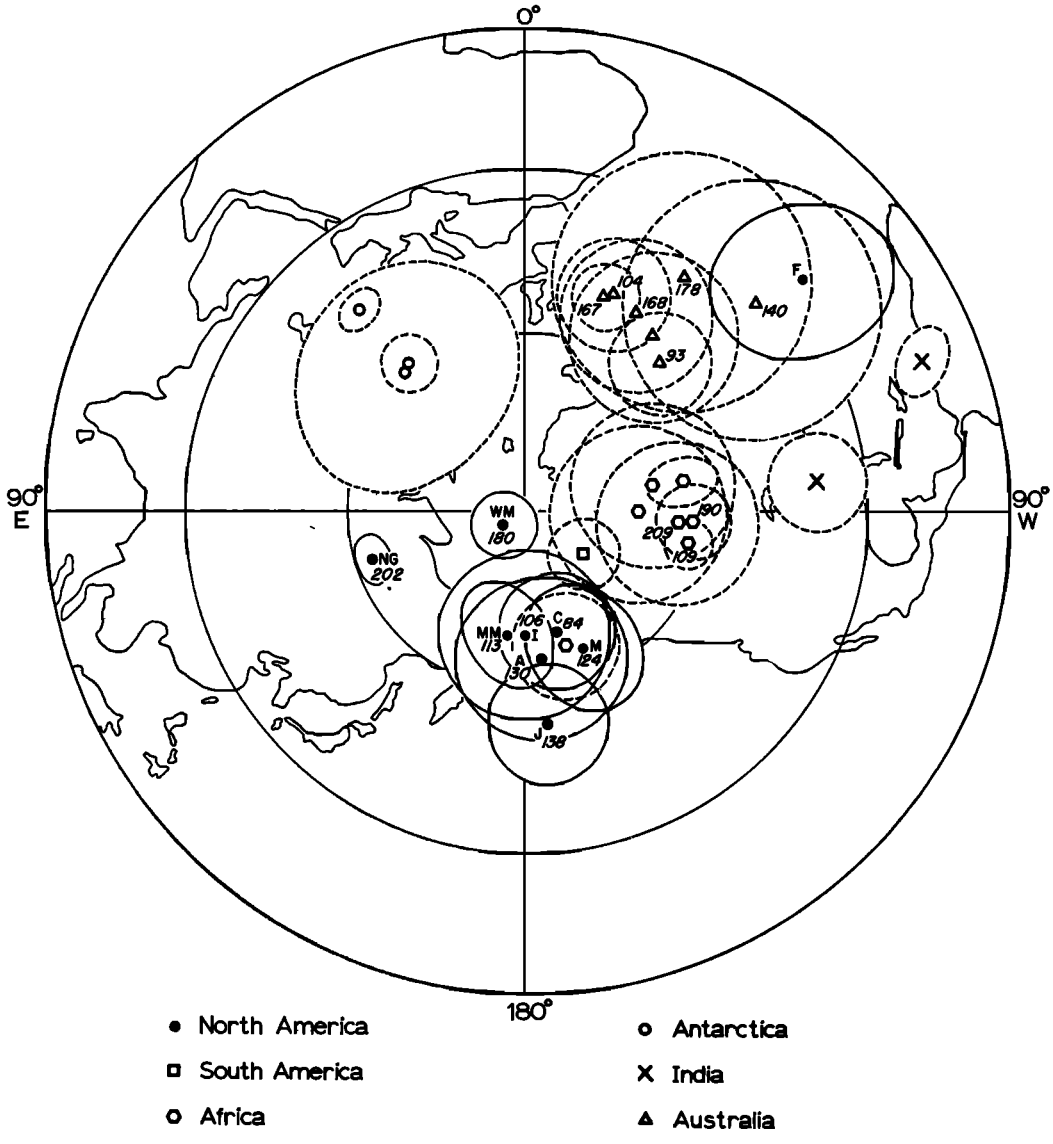



Fig. 10. Equal-area map of the northern hemisphere showing selected Mesozoic paleomagnetic pole positions for six continents. Ninety-five per cent confidence ovals or circles are shown as solid lines for North America poles and as dashed lines for others. Radiometric ages (in millions of years) are given where available; references to these ages are in the text. Poles are as follows (numbers refer to the compilation by *Irving* [1964]). North America: J, this paper; C, Sierra Nevada granites [*Grommé and Merrill*, 1965]; MM, Mount Megantic [*Larochelle*, 1961] (10.12); M, Monteregian Hills [*Larochelle*, 1962] (10.11); I, diabase near Isachsen [*Larochelle and Black*, 1963] (10.17); A, Ascutney ringdike and WM, White Mountain plutonic-volcanic series [*Opdyke and Wensink*, 1966]; F, Franciscan formation [*Grommé and Gluskoter*, 1965]; NG, Newark group [*Opdyke*, 1961]. South America: Serra Geral formation (9.53). India: Rajmahal traps (9.23); Decca traps (11.091). Antarctica: Ferrar dolerite (9.37, 9.38, 9.56). Africa: Karroo basalt (9.24); Karroo dolerite (9.27); Stormberg lavas (9.50); Lupata volcanics (10.15); Shawa ijolite [*Gough and Brock*, 1964]; Marangudzi, Mateke Hills [*Gough, Brock, Jones and Opdyke*, 1964]; Turonian rocks of Madagascar (10.08). Australia: Gibraltar microsyenite (9.28); Prospect dolerite (9.29); Tasmanian dolerite (9.30); Noosa Heads complex (9.32); Gingenbullen dolerite (9.33); Cygnet complex (10.09); Mount Dromedary complex (10.10).

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*Merrill* [1965] and supersedes the pole (10.18) listed by Irving. Two poles have been obtained from igneous intrusive rocks in Quebec; they are from the Monteregian Hills (10.11) [*Larochelle*, 1962] and Mount Megantic (10.12) [*Larochelle*, 1961]. They have been dated at 124 and 113 m.y., respectively, by *Fairbairn et al.* [1963]. A pole dated at about 130 m.y. years was obtained by *Opdyke and Wensink* [1966] from the ring-dike complex at Ascutney, Vermont. *Larochelle and Black* [1963] have determined a pole (10.17) from diabase near Isachsen on Ellef Ringnes Island in the Canadian Arctic. *Larochelle et al.* [1965] have recently redetermined the age of this rock, and the mean of the ages given is 106 m.y.

Only two other Mesozoic paleomagnetic poles are available from radiometrically dated North American rocks; they are also shown in Figure 10. One is from the White Mountain plutonic-volcanic series in New Hampshire and Vermont, dated at about 180 m.y. [*Opdyke and Wensink*, 1966]. This age corresponds to the Triassic-Jurassic boundary of *Kulp* [1961]. The other pole (8.42) is from sedimentary rocks, lavas, and diabase of the Newark group (New Jersey) of Late Triassic age [*Opdyke*, 1961]; diabase of the Palisades sill, which is part of this group, has been dated at 202 m.y. [*Erickson and Kulp*, 1961]. Other poles from the Late Triassic of North America are not radiometrically dated but are similar in position to that from the Newark group [*Irving*, 1964, p. 118]. The remaining North American pole shown in Figure 10, which lacks a radiometric age, is from the Franciscan formation of California [*Grommé and Gluskoter*, 1965].

It is clear from Figure 10 that none of the six radiometrically dated Cretaceous poles from North America is significantly different from the others. From this we can conclude that from about 138 to about 84 m.y. ago, either polar wandering relative to North America did not occur or, if it did occur, it cannot be resolved by the presently available paleomagnetic data. Before this time significant polar wandering appears to have occurred; this will be discussed further below.

The anomalous nature of the pole from the Franciscan formation is well emphasized by Figure 10. The possible causes of this nature, as discussed by *Grommé and Gluskoter* [1965], are either tectonic rotation of the rocks or a truly anomalous field direction. Clearly this pole is not explicable in terms of the other presently available late Mesozoic paleomagnetic data for North America, but we do not yet know whether the age of the Franciscan rocks is bracketed by the age range of the other poles; it could be somewhat younger.

*Evidence for axial dipole hypothesis.* The assumption, commonly made in paleomagnetism and fundamental to the preceding section, is that, when averaged over a sufficient length of time, the earth's magnetic field may be represented by a geocentric axial dipole. This assumption is based on a uniformitarian extension of the present field configuration and of paleomagnetic data from late Tertiary and Quaternary rocks; few possibilities have arisen for evaluating it for more remote times. The criteria for testing its component parts have been given by *Doell and Cox* [1961, pp. 288–289] and *Irving* [1964, p. 136]; they may be summarized as follows. If the field has axial symmetry, paleomeridians will have a common intersection. If the field is axially symmetric and geocentric, paleoisoclines drawn on the earth's surface will be concentric along the symmetry axis. If these conditions are met, the configuration of the field (e.g. dipolar, quadrupolar, etc.) can be tested directly from the spacing of the isoclines; the usual form of this test is to calculate paleomagnetic poles by means of the dipole formula and to see if they coincide. The final test, coincidence of the field symmetry axis and the earth's rotational axis, must be made by means of independent (i.e. paleogeographic) evidence [*Irving*, 1964] and is outside the scope of this paper; thus, in the present context the term 'polar wandering' refers only to motion of the geomagnetic pole.

If we allow the possibility of continental drift, tests of ancient field configuration must be made using paleomagnetic data for restricted intervals of time from widely spaced localities in single continents. The Cretaceous paleomagnetic poles for North America shown in Figure 10 fulfill these criteria in part, although they represent an appreciable segment of geologic time. These data (with the exception of the pole from the Franciscan formation) appear to favor the hypothesis that the geocentric dipole is a satisfactory approximation to the actual ancient field configuration. In the first place, the paleomerid-

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ians do not differ significantly from having a common intersection, indicating axial symmetry. This evidence is strong because the angle between the paleomeridians corresponding to the diabase on Ellef Ringnes Island and the two Sierra Nevada sites is approximately 80°, a fortunately high value. Second, that the field was substantially a geocentric dipole is suggested by the coincidence of the paleomagnetic poles calculated from the dipole formula. More data from rocks of the same age but located elsewhere in North America, especially in lower paleolatitudes, would be desirable to confirm or refute this conclusion.

*Comparison with other continents.* It is well known that the apparent polar wandering that results from the interpretation of paleomagnetic data according to the geocentric dipole model may actually be the resultant of true polar wandering and continental drift [*Irving*, 1964, p. 257]. The relative importance of these components may only be estimated by comparing paleomagnetic results from several continents; such a comparison for the late Mesozoic is illustrated in Figure 10.

The paleomagnetic poles in Figure 10 have been selected according to the following criteria (in addition to being late Mesozoic): (1) reliability according to the criteria of *Irving* [1964, p. 102], (2) availability of 95% confidence circles or ovals, (3) data obtained from igneous rocks only, and (4) ages well known, either by radiometric or stratigraphic dating (except for the pole from the Franciscan formation). Omission of data from sedimentary rocks may seem drastic, but actually of the total of about 60 independent Jurassic and Cretaceous poles listed by *Irving* [1964] only 3 of those obtained from sedimentary rocks are reliable and are accompanied by confidence intervals. The corresponding rocks are Jurassic red siltstones and radiolarities in the Alps (9.15), the Cretaceous inkstone shales of Japan (10.07), and red siltstone that is correlated with the Lupata volcanics in Mozambique (10.16). The first two of these are the only late Mesozoic poles available for Europe and Japan, respectively, and are thus isolated in space and time; the Alpine rocks are also from a tectonically disturbed area. The African pole is similar to the poles obtained from igneous rocks of probable equivalent age in the same region. Of the 33 remaining Jurassic and Cretaceous poles listed by *Irving* [1964] but not included in Figure 10, 22 are considered by him to be unreliable and 11 are either poorly dated or lack confidence intervals; these include the rest of the European and Asian poles and the 2 Jurassic poles for North America.

In addition to the North American poles discussed previously, 20 pole positions that meet the above criteria are included in Figure 10. They are from five continents: South America, India, Antarctica, Africa, and Australia. The references for these poles are given in the legend of Figure 10. Many have radiometric dates associated with them: All three antarctic poles are from the Ferrar dolerite; *McDougall* [1963] has published K-Ar ages ranging from 147 to 163 m.y. for Ferrar dolerite from various localities. The single South American pole is from the Serra Geral formation of Uruguay and Brazil; these rocks have been dated by *Creer et al.* [1965] at from 149 to 124 m.y. and more recently published ages for apparently equivalent rocks range from 147 to 119 m.y. [*Amaral et al.*, 1966; *McDougall and Ruegg*, 1966]. Six of the seven Australian poles shown in Figure 10 have been radiometrically dated, and the various ages range from 178 to 93 m.y. [*McDougall*, 1961; *Evernden and Richards*, 1962; *Irving et al.*, 1963].

Of the seven poles from the African continent that are included in Figure 10, only one, from Turonian igneous rocks of Madagascar, falls within the group of poles from North America. Two of the African poles are not listed in *Irving's* [1964] summary; they are from the Shawa ijolite in Rhodesia, dated at 209 m.y. [*Gough and Brock*, 1964], and from Marangudze and the Mateke Hills, also in Rhodesia and dated at 190 m.y. [*Gough, Brock, Jones, and Opdyke*, 1964]. The Lupata volcanics (pole 10.15) have been dated at 109 m.y. [*Gough, Opdyke, and McElhinny*, 1964]. Three of the closely grouped African poles for which no ages are shown in Figure 10 are from rocks designated as part of, or correlated with, the Karroo system; *McDougall* [1963] has published ages for Karroo dolerites ranging from 190 to 154 m.y. Thus, at least two and possibly more of the African poles included in Figure 10 are as old as Late Triassic.

It is apparent from Figure 10 that for Africa and Australia, as well as North America, the

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paleomagnetic poles representing large and overlapping segments of Mesozoic time are so closely grouped for each continent, that differences between individual poles cannot be said to indicate polar wandering. Moreover, there are striking separations among the three groups of poles. The stability of paleomagnetic poles relative to Africa has been pointed out by *Gough, Opdyke, and McElhinny* [1964], who find that significant polar movement occurred between Early Permian and Late Triassic time and again since about the middle of the Cretaceous. Relative to Australia, however, pole positions actually show little or no change from Late Carboniferous until early Tertiary time [*Irving et al.*, 1963; *Irving*, 1964, p. 139, 1966]. We have shown above that any possible change in the pole position for North America between latest Jurassic and middle Late Cretaceous time was too small to be detected with presently available data, but a large change in the North American pole position apparently occurred between the Late Triassic and the end of the Jurassic (see *Irving* [1964, p. 138] and Figure 10 of this paper). The data from Antarctica and South America do not cover sufficiently large spans of time to allow any similar conclusions to be drawn from them. Finally, two paleomagnetic poles for India (from the Jurassic Rajmahal traps and the Late Cretaceous or Eocene Deccan traps) have also been included in Figure 10. They indicate considerable apparent polar movement relative to India, but the Deccan traps are probably younger than any of the other rocks represented in the figure.

We conclude from the data shown in Figure 10 that during the Jurassic and much of the Cretaceous the rate of true polar wandering was either zero or too small to be resolved paleomagnetically. It seems highly improbable that during Late Carboniferous, Permian, and Triassic time polar movement relative to Australia was exactly compensated by drifting of that continent; we therefore can extend this conclusion to cover at least the span of time between the beginning of the Permian and the middle of the Cretaceous periods. On the other hand, the times of relatively rapid apparent polar wandering for Africa, Australia, and North America are not concurrent, as pointed out above. It follows that the apparent polar wandering relative to North America and Africa, during the Jurassic and Triassic, respectively, is due predominantly or entirely to continental drift.

## Summary and Conclusions

NRM directions have been determined at thirteen sampling sites in two bodies of intrusive igneous rock in the western Sierra Nevada. Both reversed and nomal directions of NRM are found in both plutons, and in some areas closely adjacent specimens have opposing polarities. Soft components of NRM have been removed by ac demagnetization of representative groups of specimens from all sites. Optimum demagnetizing fields for the various sites range from 75 to 800 oe. The site mean NRM directions after demagnetization of all specimens (shown in Figure 9) appear to have a non-Fisherian distribution, and this could be due to a nonuniform distribution of sampling sites with respect to time. Thus, it is not altogether clear whether sufficient time is represented by these sites to average out the effect of secular variation in the calculation of the mean paleomagnetic pole. Nevertheless, any systematic bias of the mean pole (shown in Figure 10) is probably negligible compared with its 95% confidence cone ($\alpha_{95} = 9.5°$).

Including the one determined in the present work, six paleomagnetic pole positions are now available for North America from radiometrically dated igneous rocks that range in age from 138 to 84 m.y. This interval of over 50 m.y. corresponds to the first two-thirds of the Cretaceous period. None of the six pole positions can be said to be significantly different from any of the others (Figure 10), from which we conclude either that apparent polar wandering relative to North America did not occur during that time or that its rate was too slow to be detected.

The fact that these six Cretaceous North American poles are derived from rocks in three widely separated localities provides new evidence that the earth's field was a geocentric dipole during that time (just as it clearly has been during the latter part of Cenozoic time). In particular, the angle between the paleomeridians for the California and Quebec sites is about 45°, and the corresponding angle between the California and Ellef Ringnes Island sites is close to 80°. That these three widely sep-

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arated paleomeridians for a single continent have essentially a common intersection constitutes exceptionally strong evidence for axial symmetry of the geomagnetic field. Furthermore, calculating the paleomagnetic poles by means of the dipole formula clearly produces very good correspondence among the six sets of data.

An unequivocal interpretation of the aberrant paleomagnetic pole from the Franciscan formation of California (Figure 10) is no closer at present. The age of these rocks is not well known, but additional paleomagnetic data from well-dated Late Cretaceous and early Tertiary rocks of North America might eventually provide such an interpretation.

The stability of paleomagnetic pole positions for North America during most of Cretaceous is characteristic of at least two southern hemisphere continents for even longer periods: poles determined from Australia show very little shift from Late Carboniferous until early Tertiary time, and poles from the main continent of Africa show essentially no shift from Late Triassic up to middle Cretaceous time. On the other hand, a large polar shift relative to Africa seems to have occurred between Early Permian and Late Triassic time, and an even larger and possibly more rapid shift occurred relative to North America during the Late Triassic and Jurassic. Because the various periods of rapid apparent polar movement relative to these two continents clearly did not coincide in time and during the same time the Australian pole position remained essentially constant, we conclude that, during the Mesozoic at least, apparent polar wandering was due almost entirely to continental drift, and true polar wandering either did not occur or was greatly subordinate. It remains to be seen whether this conclusion applies to other segments of geologic time.

It is also now clear, from this work and that of *Irving* [1966], that apparent polar wandering, hence by implication continental drift, is an intermittent process. Rates of apparent polar wandering cannot be translated directly into rates of continental drift, because of the indeterminacy of paleolongitude. The general pattern for continents considered singly seems nevertheless to be long periods of essentially no drift, preceded and followed by shorter periods of relatively rapid drift.

## Appendix

Locations of samples for which potassium-argon ages are given in Table 1 are given below.

*KA 1432, KA 1433.* Approximately 2½ miles southwest of site 12 in central lobe of Bucks batholith. Center of south ½ of southwest quarter section 30, township 24 north, range 6 east, Pulga 15′ quadrangle.

*KA 1436, KA 1437.* At site 10, in outer zone of northeast lobe of Bucks batholith, ½ mile north of Workman's Bar. Center of north half section 2, township 24 north, range 6 east, Pulga 15′ quadrangle.

*KA 1438, KA 1439.* At site 4, near center of Merrimac batholith, 1 mile southwest of Merrimac near Coon Creek. Center of west ½ of southwest quarter section 16, township 22 north, range 6 east, Pulga 15′ quadrangle.

*KA 1440, KA 1441.* Near southeast margin of western lobe of Bucks batholith, ¼ mile north of Cresta powerhouse. Center of east ½ of east ½ of southwest quarter section 22, township 23 north, range 5 east, Pulga 15′ quadrangle.

*KA 1442, KA 1449.* At site 8, in hornblende diorite adjacent to outer zone of northeast lobe of Bucks batholith, at Workman's Bar. Center section 2, township 24 north, range 6 east, Pulga 15′ quadrangle.

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